Latent Semantic Indexing

Final Project for the Information Retrieval Course

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Our Workflow

1. Data Preprocessing and Term Document Matrix

Creating a term-document matrix, where each row represents a unique term, each column represents a document, and each cell contains the frequency (or TF-IDF score) of the term in the document. Data must be handled and preprocessed.

2. SVD and LSI

Applying Singular Value Decomposition (SVD) to the TD matrix. SVD reduces the high-dimensional term space into a lower-dimensional semantic space, where similar words and documents are grouped closer together based on their usage patterns—not just their exact terms. This is the core in LSI.

3. IR System and Query handling

Mapping both documents and queries into this semantic space, allowing the system to find relevant results even when the query and the documents don’t share exact keywords. Queries must be preprocessed.



Data Handling

In this project we used a CRAN dataset imported from kaggle, which contains documents in a dot-tagged format, encoded as follows:

- .I → Document ID
- .T → Title
- .A → Author(s)
- .B → Bibliographic info
- .W → Abstract



Data Preprocessing

We decided to allow for a dynamic selection of preprocessing operations to be applied to the text:

The wizard summoned a firestorm that scorched the battlefield for 18 damage.

LOWERCASE

the wizard summoned a firestorm that scorched the battlefield for 18 damage.

PUNTUATION

the wizard summoned a firestorm that scorched the battlefield for 18 damage

STOP WORDS

wizard summoned firestorm scorched battlefield 18 damage

LEMMATIZATION

wizard summon firestorm scorch battlefield 18 damage

NUMERIC TOKENS

wizard summon firestorm scorch battlefield damage

NOUN VERB NOUN VERB NOUN NOUN **PART OF SPEECH (POS)**



Term Document Matrix Boolean



	doc 1	doc 2
term 1	0	0
term 2	1	1
term 3	1	0

Each entry is just the boolean value about the presence of term_i in document j. It’s the simplest representation.



Term Document Matrix TF



	doc 1	doc 2
term 1	3	0
term 2	1	2
term 3	2	0

Each entry of the matrix contains the relative term frequency (how many times term_i appears in document_j). It is still a sparse matrix and the time-cost of computation is similar to the boolean version



Term Document Matrix TF-IDF

	doc 1	doc 2
term 1	0.549	0,997
term 2	0.334	0.835
term 3	...	...

TF-IDF is a measure used to evaluate how important a word is to a document relative to a collection of documents. It increases with the number of times a word appears in a document but is offset by how frequently the word appears across all documents, highlighting more informative terms.

$$idf_t = log(\frac{N}{df_t})$$

$$tf\text{-}idf_{t,d} = tf_{t,d} \times idf_t$$



SVD Core Concept

SVD (Singular Value Decomposition) factorizes a matrix M into three components: V, U and Σ where Σ contains the singular values. It allows us approximate our matrix with using one with lower rank.
For our purpose it is useful to represent documents in a more convenient vector space with lower dimension and linear combinations of terms as a basis.

$$U\Sigma V^T$$

Term-Concept matrix
The matrix which diagonalizes AAt with orthonormal eigenvectors of AAt as columns

Singular Values Matrix
The diagonal matrix containig singular values

$$\sigma_i = \sqrt{\lambda_i}$$

where λi are the eigenvalues of AtA and AAt

Document-Concept matrix
The matrix which diagonalizes AtA with orthonormal eigenvectors of AtA as columns



LSI Explained

$$U_k \Sigma_k V_k^T$$

Term-Concept matrix
Each column represents how much each term is represented by a certain concept

Concept matrix
Each value represents the "weight" of a concept

Document-Concept matrix
Each column represents how much each term is represented by a certain concept

k is the reduced dimension



Query Handling

The free form text query is handled just like te documents:

- As a first step, the query is passed trough the same preprocessing pipeline used to clean the documents
- The cleaned query is then turned into its term vector representation, either with boolean or term frequency encoding



IR System : Bringing all together

Our Python Implementation

One can run the notebook on a toy dataset or use advanced features in the Python file to try the IR on more complex collections.



Thank you!

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